

Investigating the classical regime of Incompressible Schrödinger Flow

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ABSTRACT

Modelling and conserving vortex filaments in a time-efficient manner in fluid simulations has long been a challenge. The publication "Schrödinger Smoke" introduced a novel method that utilises spherical Clebsch variables to conserve vortex filaments in a grid-based model, combining the advantages of Lagrangian and Eulerian methods despite its model relying on propagation via the Schrödinger equation on a grid-based domain. In this report, we tested their model against an Eulerian equivalent in addition to investigating its Classical limit $\hbar \rightarrow 0$. Our results support the Incompressible Schrödinger flow model as being more accurate and time efficient in replicating the physical leapfrog phenomena than equivalents based on Euler equations. Initial tests suggest a complexity scaling relation with $O(\log(x))$ within the tested regime, however, context suggests this is likely erroneous. Results suggest that reducing $\hbar$ causes a loss of information, indicating that the Schrödinger Smoke model fails to retain its advantages in the classical regime.

Key words: methods: numerical, fluid simulation, Schrödinger equation, vorticity confinement

1 INTRODUCTION

The Incompressible Schrödinger flow (ISF) is a fluid simulation model published by [Chern et al. \(2016\)](#). It utilises a two-component solution to the modified Schrödinger equation $i\hbar\psi_t = -\frac{\hbar^2\nabla^2\psi}{2} + p\psi$ on a periodic 3D grid to model fluid dynamics. In this report, we replicate the model and investigate some of the questions posed at the end of the Schrödingers Smoke paper regarding whether ISF approaches the Euler equations in its classical limit, $\hbar \rightarrow 0$. Furthermore, to extend on the comparison between ISF and Euler equation-based model we developed an Euler equation equivalent, using the same operator splitting, order of operations and FFT-based Poisson solvers to facilitate a fair comparison between their algorithms and run times.

This report aims to further discuss the physical applications of ISF in addition to its predecessors and a recent publication by [Yang et al. \(2021\)](#) that takes an alternate approach of directly modifying the Euler equations to achieve similar results. The ISF equation is originally derived from the non-linear Gross–Pitaevskii (GP) Schrödinger equation, $i\hbar\phi_t = -\hbar^2\nabla^2\phi + \frac{1}{b^2}(|\phi|^2 - 1)\phi$. The GP equations describe the ground state of a system of identical bosons under the Hartree–Fock approximation, which can be interpreted as a dilute approximation where the scattering length exceeds the spacing between particles. This has led to applications in not only Bose-Einstein condensates but also a diverse set of fields including wave-like dark matter models like axions ([Glennon et al. 2022](#)) and super fluids([Okulov 2012](#)).

ISF shows particular similarities to the superfluid models, whereby the cubic term in the GP equation, opposes deviations from $\phi = 1$, resulting in an incompressible fluid, $\phi = 1$ and $\nabla \cdot \mathbf{u} = 0$, as $b \rightarrow 0$. This allowed the Schrödinger’s smoke model to substitute the non-linear cubic term with a linear Lagrange multiplier, $\nabla \cdot \mathbf{u} - \nabla^2 p = 0$, limiting the model to the incompressible regime but simplifying and accelerating the required algorithms in the process. Even when limited to the incompressible regime the GP equations remain applicable as shown in publications on two-component Bose-Einstein condensates([Lee & Lin 2008](#)) and superfluid-classical fluid interactions ([Brachet et al. 2023](#)).

Section 2 will discuss the mathematical background in addition to introducing the algorithms applied. Following this, the results are presented in Section 3 and discussed in further detail in Section 4. Section 4 includes details regarding code optimisation, bottlenecks and the physical regimes that were investigated. Section 5 concludes the report. All .mp4 video files associated with Figure images are named according to their associated Figures, meaning that when referring to Figure 3c, reference is also made to the associated Figure 3c.mp4.

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2 BACKGROUND

2.1 Modelling vorticity

Schrödinger's Smoke exemplifies the potential of the ISF simulations by their applicability in visual simulations. They demonstrate how systems with complex vortex filaments, whether it be mist around an obstacle or rings performing consecutive leapfrogs, can be faithfully reproduced. Allowing ISF to compete with state-of-the-art Euler equation-based models at comparatively low resolutions, on time scales of around half an hour on a laptop.

This makes ISF particularly promising as modelling vortex elements has remained a consistent challenge as grid-based models tend to lose the information between grid cells, leading to numerical diffusion. This hampers the ability to replicate observations in nature, like a bubble or smoke vortex rings, where vortex elements and filaments can remain stable for prolonged periods (Naitoh et al. 1995)(Maxworthy 1972). To combat this loss of information in fluid simulations various approaches have been developed. Lagrangian methods treat the vortex elements as a discrete set of particles that are advected along the fluid, however, these result in difficulties relating to the initialisation and later formation of new vortex elements and replicating linking of vortex filaments, requiring them to be artificially connected (Leonard 1980)(Angelidis & Neyret 2005). This, combined with unwanted particle accumulation and dispersion phenomena can lead to artefacts in the simulation (Rossinelli & Koumoutsakos 2008). Lagrangian methods also model them as individual particles, which becomes computationally expensive when trying to model millions of particles. Further This led to the introduction of hybrid methods, that apply the Lagrangian vortex particles whilst maintaining an underlying grid based on Euler equations. Although hybrid models are widely used today (Christiansen 1973)(van Rees et al. 2011)(Kudela & Kozłowski 2009) and overcome many of the issues with the Lagrangian approach, interaction between the grid and particle frameworks introduced new challenges, with a particular issue being artefacts near boundary conditions (Anderson & Reider 1994). This provides one of the major advantages of ISF, preserving vortex filaments on a faster grid based model without additional modifications or post-processing. A further review of the history of modelling vortex elements can be found in Mimeau & Mortazavi (2021).

2.2 Mathematical background

What allows ISF to conserve its vortex elements without a Lagrangian or hybrid-based approach are the Clebsch variables used to define the system. Clebsch variables re-parameterise the velocity vector field in terms of scalar fields (Clebsch 1857) (Clebsch 1859). For example one can define $\mathbf{u} = \nabla A$, however when determining the vorticity of the system we find that due to the curl of a gradient being zero, the field becomes irrotational, $\omega = \nabla \times \nabla A = 0$. To retain vorticity, a minimum of two scalar fields are required, $\mathbf{u} = B\nabla C$. Allowing the vorticity to be non-zero, $\omega = \nabla \times (B\nabla C) = \nabla B \times \nabla C$. This leads us to the original Clebsch variables, $\mathbf{u} = \nabla A + B\nabla C$, where we define $\{A, B, C\} = \phi : M \rightarrow \mathbb{R}^3$ and similarly for the vorticity, $\{B, C\} = s : M \rightarrow \mathbb{R}^2$. However, as noted and derived in the original ISF paper, this Clebsch map has a vanishing helicity. Helicity, $H = \int_M \mathbf{u} \times \omega$, describes the chirality of the system and plays an important role in defining the knottedness and chirality of vortex filaments.

To overcome this limitation spherical Clebsch maps were introduced, which instead resulted in quantised helicities of $\omega = 2\pi\hbar n$ (for proof see Appendix D in Chern et al. (2017)). Although inconvenient in Classical models, in quantum systems this is less of an issue as quantum superfluids naturally exhibit quantised vortices through independent mechanisms (Hall & Vinen 1956). The resulting Clebsch maps have the following form $\phi : M \rightarrow \mathbb{S}^3$ and $s : M \rightarrow \mathbb{S}^2$.

It is convenient to represent these as manifolds embedded in a 4D space. Such a system can either be defined in terms of a quaternion coordinate system, $H = a + ib + jc + kd$, or an equivalent two-component complex field $(a + ib, c + id) = \psi : M \rightarrow \mathbb{C}^2$. This is where we derive the two-component wave function, as a solution to the Schrödinger equation. The spherical manifolds arise from the incompressibility constraint, $|\psi|^2 = 1$, which reduces the system to the surface of a 3-sphere, $\mathbb{S}^3$. The $\mathbb{S}^2$ Clebsch map representing the vorticity of the system can then be determined using a Hopf map (Hopf 1931) which projects from $\mathbb{S}^3 \rightarrow \mathbb{S}^2$, where $\mathbf{s} = [a^2 + b^2 - c^2 - d^2, 2(bc - ad), 2(ac - bd)]^T$. This allows us to determine the energy of the system, $H_{IMF} = \frac{\hbar^2}{2} |d\psi|^2 = \frac{1}{2} |v|^2 + \frac{\hbar^2}{8} |ds|^2 = H_e + H_s$, where H_e represents the kinetic energy as would be derived in the Euler equations and H_s represents the Dirichlet energy which in rotational fluids can be associated with the combined length of all vortex filaments in the system. This results in an important contribution whereby the energy of the vortex filaments is now conserved with time, which gives ISF the edge in conserving such elements.

Projecting from $\mathbb{C}^2$ to $\mathbb{S}^3$ leaves an unconstrained global phase that can be associated with a gauge transformation $\psi \rightarrow e^{i\frac{p}{\hbar}}\psi$. In terms of velocity this gauge results in $\mathbf{u} \rightarrow \mathbf{u} + \nabla p$, noting that ∇p is irrotational. By gauge fixing $\nabla \cdot \mathbf{u} = 0$, we extract the Poisson equation $\nabla^2 p = \nabla \cdot \mathbf{u}$, which corresponds to using p as a Lagrange multiplier to apply the incompressibility condition. For a more in-depth and rigorous analysis of spherical Clebsch variables and the underlying differential geometry, we refer the reader to Chern et al. (2017).

2.3 Mandelung transforms

The validity of modelling fluids using Schrödinger's equation is not immediately clear. However such transforms fall under the category of Mandelung transforms (Mandelung 1927) which is a well-established field. We define the flow velocity in terms of the probability current, $\frac{\rho(x,t)\mathbf{u}}{m} = \mathbf{j} = \frac{-i\hbar}{2m} (\bar{\psi}\nabla\psi - \psi\nabla\bar{\psi})$. By then inserting $\psi(x,t) = \sqrt{\frac{\rho(x,t)}{m}} e^{iS(x,t)}$ into the free particle Schrödinger equation we get, $\mathbf{u} = \hbar\nabla S(x,t)$, with S representing the phase of $\psi = \psi_1 + \psi_2$. Note that we set $m=1$, and based on the incompressible constraint we can define $|\psi|^2 = \rho = 1$. This allows us to determine the velocity vector field of the grid in terms of ψ , which we then use to solve for the Lagrange multiplier p in terms of $\nabla^2 p = \nabla \cdot \mathbf{u}$.

Although the Mandelung transforms have only been robustly confirmed for 1D and 2D (Lin & Xin 1999)(Jerrard & Spirn 2013), according to the Schrödinger's Smoke paper they are expected to hold for 3D. To further investigate this, an Euler equation simulation was developed based on the building blocks of the ISF code, ensuring identical order of operations, FFTs, Poisson solvers, etc. This allowed for comparisons between the models with minimal differences which enabled more in-depth testing of relative run times, performance at equal and difference resolutions, and a comparison of how

well vortex elements are preserved with time. In particular, a model of two initialised vortex rings leapfrogging, termed the 'leapfrog' model, was used as a benchmark for determining how many leapfrog cycles a simulation and set of parameters could maintain the two distinct vortex rings. By quantifying the number of times a vortex 'leaps' around the other, a pseudo-quantitative method of comparing models was established. This provides an alternative to the limited number of analytic solutions to 3D vortex-based systems via the Euler equations.

2.4 Applications in the Classical regime

Despite the original application in a quantum mechanical framework, a publication following the work of Schrödinger's Smoke, by Yang et al. (2021) applies the same spherical Clebsch map representation to Euler equations to model purely classical flow, opening the possibility for applications in aerodynamics where vorticity can have a major impact on the drag of aircraft like planes and helicopters (Zhu et al. 2002)(Shi et al. 2020)(Kudela & Kozłowski 2009). This also reintroduces the question regarding the classical limit of the ISF model as $\hbar$ results in quantised vortex filament radii of $2\pi\hbar$. Whether the conservation of vortices and formation of aggregate vortexes holds as $\hbar \rightarrow 0$ is unclear from existing work, prompting further investigation.

3 ALGORITHMS

Algorithm 1 Loop for incompressible Schrödinger flow

Input: $\psi^{init}, dt, \hbar$

- 1: **for** $t = 0$ to t_{max} , $stepsize = dt$ **do**
- 2: $\psi \leftarrow \text{SCHRÖDINGER_FLOW}(\psi)$
- 3: $\psi \leftarrow \psi/|\psi|$
- 4: $\psi \leftarrow \text{VELOCITY_CONSTRAINT}(\psi)$
- 5: $\psi^{new} \leftarrow \text{PRESSURE_PROJECT}(\psi)$
- 6: $particle_x \leftarrow \text{STAGGERED_ADVECT}(grid_v, particle_x)$
- 7: **end for**

=0

The ISF algorithm outlines the main feedback loop that the algorithm is based on. The Schrödinger equation is solved using operator splitting. This is achieved by first solving the kinetic energy ODE in Fourier space using Algorithm 2, then normalising ψ . Algorithm 3 then applies any potentials in the form of velocity constraints, followed by Algorithm 4 which applies the Lagrange multiplier in the Fourier domain to enforce the incompressibility constraint. The resultant wave function is then used to re-calculate the velocity and advect tracer particles on the grid using interpolation. These algorithms were run using a 12-core i7-1260P Intel CPU and for the parameters shown in Figure 1 took 18 minutes to complete. (Note that when unaccompanied by y and z: x represents a component-wise operation).

Algorithm 2 FFT time integration of Schrödinger equation

function $\text{SCHRÖDINGER_FLOW}(\psi) \leftarrow \text{FFT3D}(\psi)$
 $k_x \leftarrow 2\pi \frac{(n_x - \frac{1}{2})}{dx}$ # k-space grid $\lambda \leftarrow \frac{\hbar}{2} |\mathbf{k}|^2$
 # solving free particle SE $\tilde{\psi} \leftarrow e^{-i\lambda dt} \tilde{\psi}$ $\psi \leftarrow \text{INVFFT3D}(\tilde{\psi})$ **return** ψ

Algorithm 2 Solves the Schrödinger differential equation, excluding the pressure component, in the Fourier domain and multiplies it as a phase to $\tilde{\psi}$ which is then inverse transformed and returned.

Algorithm 3 Apply velocity or potential to a particular region

function $\text{VELOCITY_CONSTRAINT}(\psi)$

- 1: **if** velocity constraint **then**
- 2: $\mathbf{k}_v \leftarrow \frac{\mathbf{v}}{\hbar}$
- 3: $\omega_v \leftarrow \frac{|\mathbf{v}|^2}{2\hbar}$
- 4: $\psi|_{\Omega} \leftarrow e^{i(\mathbf{k}_v \mathbf{x} - \omega_v dt)} |\psi|$
- 5: **else if** potential **then**
- 6: $\mathbf{k}_{pot} \leftarrow \frac{\sqrt{2U_{pot}}}{\hbar}$
- 7: $\omega_{pot} \leftarrow \frac{U_{pot}}{\hbar}$
- 8: $\psi \leftarrow e^{i(\mathbf{k}_{pot} \mathbf{x} - \omega_{pot} dt)} |\psi|$
- 9: **end if**
- 10: **return** ψ

Algorithm 3 Applies a potential to the wave function as either a velocity constraint $\mathbf{v}$ or a potential U_{pot} . Velocity constraints are applied by enforcing a phase on a certain region Ω . Potentials, like gravity, are approximated as plane wave phase shifts which multiply with ψ .

Algorithm 4 Solve Poisson equation to ensure divergence free constraint

function $\text{PRESSURE_PROJECT}(\psi)$ $\mathbf{v}_x \leftarrow \frac{\hbar}{i} \cdot \text{ANGLE}(\langle \psi^x | \psi^{x-1} \rangle)$ # $\hbar \nabla S(x, t)$ $d\mathbf{v} \leftarrow \nabla \cdot \mathbf{v}$
 $\tilde{d\mathbf{v}} \leftarrow \text{FFT3D}(d\mathbf{v})$ $k_x \leftarrow \frac{\sin(2\pi n_x)}{dx}$ # k-space
 grid $\tilde{p} \leftarrow \frac{\tilde{d\mathbf{v}}}{-|\mathbf{k}|^2}$ # solving Lagrange multiplier
 $p \leftarrow \text{INVFFT3D}(\tilde{p})_{\mathbb{R}}$ **return** $e^{-\frac{ip}{\hbar}} \psi$

Algorithm 4 Derives a velocity based on the phase difference between ψ^x and ψ^{x-1} it's neighbouring grid location. The differential of $\mathbf{v}$ is then transformed to the Fourier domain where $d\mathbf{v}$ is scaled by the global phase at each grid location, μ . This is then FFTed back as a phase shift which is applied as the Lagrange multiplier to enforce the incompressible constraint.

Algorithm 5 Advect tracer particles along grid

function $\text{STAGGERED_ADVECT}(grid, particle)$

- 1: $w_{x,1} \leftarrow \text{scipy.interpn}(grid_x, grid_v, particle_x)$
- 2: $w_{x,2} \leftarrow \text{scipy.interpn}(grid_x, grid_v, particle_x + \frac{w_{x,1} dt}{2})$
- 3: $w_{x,3} \leftarrow \text{scipy.interpn}(grid_x, grid_v, particle_x + \frac{w_{x,2} dt}{2})$
- 4: $w_{x,4} \leftarrow \text{scipy.interpn}(grid_x, grid_v, particle_x + w_{x,3} dt)$
- 5: $particle_x \leftarrow particle_x + \frac{dt}{6} (w_{x,1} + 2w_{x,2} + 2w_{x,3} + w_{x,4})$
- 6: **return** $particle_x$

Algorithm 5 Determines the change in particle positions, $particle_x$, using a weighted mean of the interpolated velocities, using Scipy's trilinear interpolation package (Virtanen et al. 2020), according to the Runge-Kutta 4th order method (Runge 1895) (Kutta 1901), resulting in an error of order $O(dt^4)$.

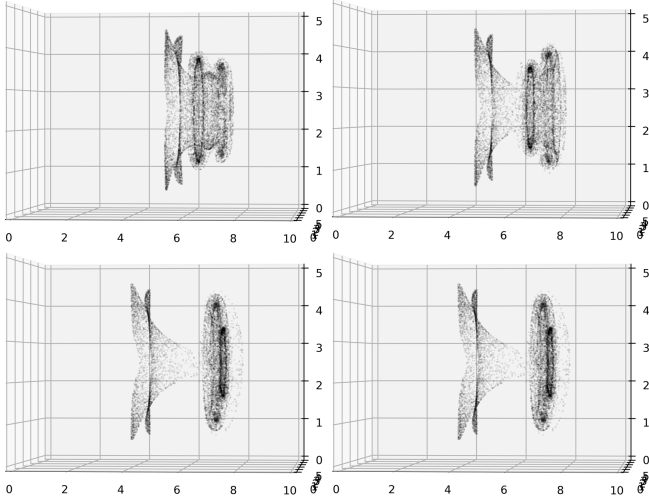


Figure 1. Four sequential time steps. Illustrating the leapfrog process where the inner ring is passing through the outer ring, then starts expanding to become the new outer ring. Parameters: $\hbar = 0.1$, $dt = 0.3$, $tmax = 25$, $v_x = -0.2$, $res = [128, 64, 64]$, $gridsize = [10, 5, 5]$.

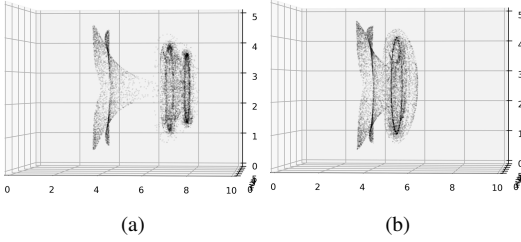


Figure 2. Comparing ISF (a) and Euler (b) after 1 "leap". ISF shows two distinct loops after having undergone a leap, whilst Euler fails to complete any leaps before becoming indistinguishable. Parameters: $\hbar = 0.1$, $dt = 0.03$, $v_x = -0.2$, $res = [128, 64, 64]$.

4 RESULTS

4.1 Model comparison

Several models in Dr Chern's paper were successfully replicated, including the leapfrog, jet and gravitational potential models. This allowed us to compare leapfrog simulations, reaching eight leaps as seen in **Figure 1**, in addition to making an Euler equation equivalent to the leapfrog simulation. As seen in **Figure 2** despite identical parameters the Euler model failed to produce coherent inner and outer rings so several parameters were tested to determine whether a higher resolution, **Figure 2a**, or smaller timesteps, **Figure 2b**, could recover the leapfrog phenomena. However, the only modification that showed distinguishable rings and resulted in two leaps was reducing the radius of the inner ring by a factor of 0.55.

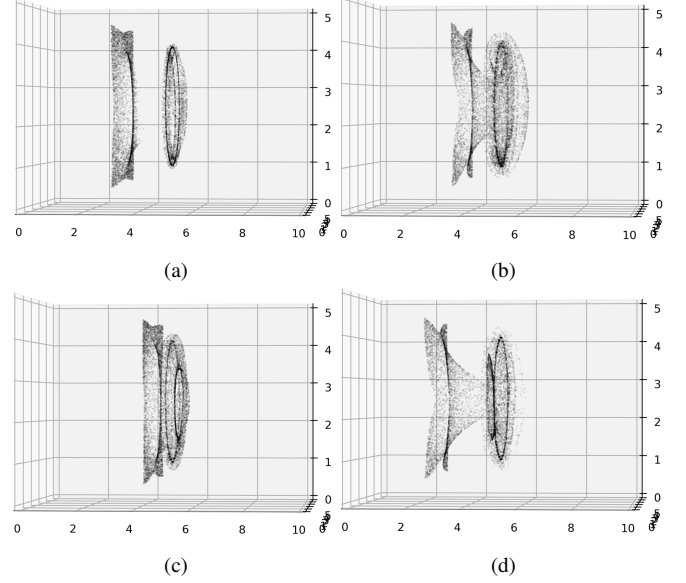


Figure 3. Euler with different parameter modifications. Comparing Euler with doubled resolution(a), 0.25 times smaller dt (b), and reducing the radius of the inner loop to $r=0.5$ from $r=0.9$ (c,d).

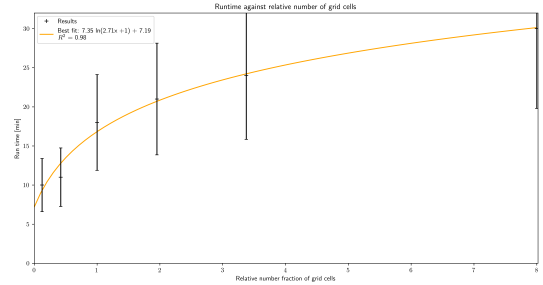


Figure 4. Run time against relative grid cell count. Shows a preliminary fit to a $\log(x)$ based function, indicates $O(\log(x))$ within parameter space tested. Error bars of 34% are applied due to singular measurements resulting in high uncertainties. Constant parameters: $\hbar = 0.1$, $dt = 0.03$, $v_x = -0.2$.

4.2 Computational efficiency

The Euler model with double the resolution of the ISF leapfrog takes 25 minutes to run compared to 18 minutes with the ISF, making it 38% slower despite failing to resolve a single leap, seen in **Figure 3**. Even if the Euler model were to succeed at even higher resolutions, it would still be less time-efficient at modelling the same physical phenomena to an equivalent physical accuracy. The caveat of potentially different run time scaling applies, however considering their algorithms are similar, in particular the biggest bottlenecks of FFT and interpolation, this is less likely. To determine the scaling of run-time against the resolution of the ISF model the resolution was scaled by factors of 0.25 compared to the original and timed (equating to changes in the total number of grid cells by $(1 + 0.25n)^3$). The resulting plot in **Figure 4** suggests a big O notation of $O(\ln(x))$ within the tested regime. It is however important to note the observational bias whereby at higher resolutions run times become unfeasible, so higher-order terms might have been missed.

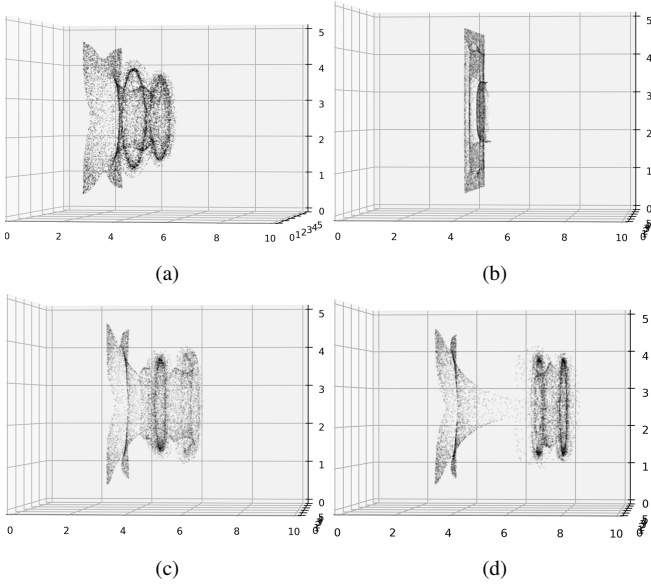


Figure 5. Comparing ISF with the following scaled parameters: $\frac{1}{2}\hbar$ (a), $\frac{1}{10}\hbar$ (b), $(\frac{1}{2}\hbar, \frac{1}{2}v_x)$ (c), $(\frac{1}{1000}\hbar, \frac{1}{1000}v_x, \frac{1000}{1}dt)$ (d).

4.3 Testing the classical regime

In investigating how ISF behaves in the classical limit, reductions in $\hbar$ rapidly lead to the loss of information as seen in **Figure 5b**. However by downscaling the velocity by the same factor as $\hbar$ the evolution of the system appears to slow down significantly, but retains its structure, as seen in **Figure 5c** and **5d**.

To understand this we need to look at the energy equation of the system $H_{ISF} = H_e + H_s = \frac{1}{2}|v|^2 + \frac{\hbar^2}{8}|ds|^2$. Here it becomes clear that by scaling $\hbar$ and v by the same factor the ratio between the kinetic energy, H_e , and the Dirichlet energy, H_s , is conserved. This can be seen in **Figure 6**, where scaling $\hbar$ on its causes a change in the relative energies, whilst scaling $\hbar$ and v preserves it.

To counterbalance the reduced velocity the time total and stepsizes can be increased, as seen in **Figure 5d** this also appears to preserve the structure, allowing it to perform several leaps, about 4. However further investigation is required to determine the validity of scaling dt . Increasing dt without $\hbar$ quickly leads to loss of information as large time steps increase the numerical error in each step.

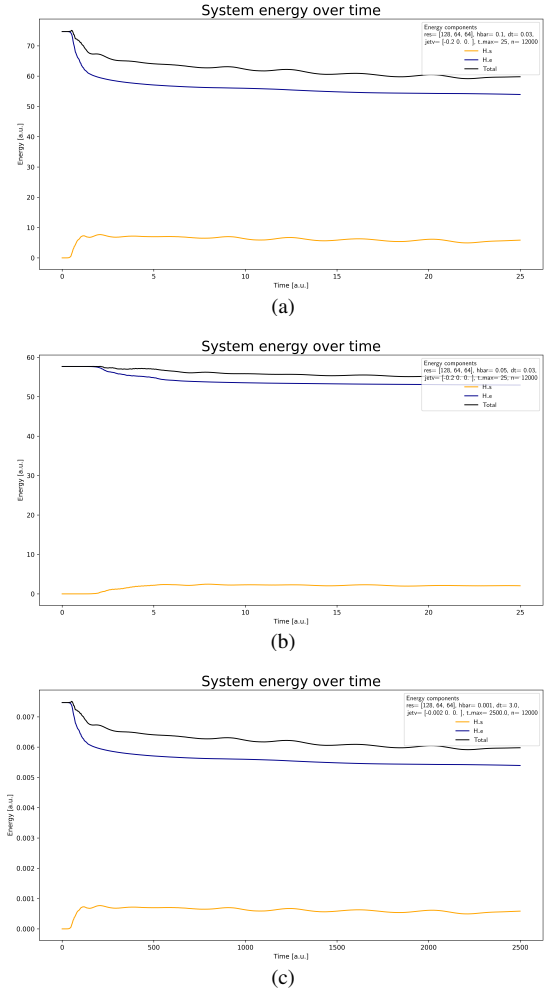


Figure 6. Energy over time. H_e and H_s for the following scaled parameters: no scaling (a) $\frac{1}{2}\hbar$ (b) $(\frac{1}{2}\hbar, \frac{1}{2}v_x, \frac{2}{1}dt)$ (c). The y axis represents energy in arbitrary units.

5 DISCUSSION

5.1 Applicability to the classical regime

The results in **Figure 3** suggest that the ISF model outperforms it's Euler equivalent in modelling the physical leapfrog phenomena despite requiring only half the resolution. As pointed out in the Schrödinger's Smoke paper this held for state-of-the-art Euler-Lagrangian hybrid models when published. As the derived spherical Clebsch maps are classical in nature, the prospect of achieving the vortex filament resolution of the ISF model in a classical fluid model would have significant implications for modelling fluids in a variety of highly rotational physical systems, whether it be aeroplane drag, weather prediction or modelling astrophysical systems like black hole accretion disks.

This likely motivated the question, at the end of Schrödinger's Smoke, of whether the ISF model in the classical limit, $\hbar \rightarrow 0$, results in the Euler equations, as this would suggest compatibility of applying it to classical systems. **Figure 5** suggests that this is not the case, but implies that by also scaling the initial velocity and time components with $\hbar$ that the system dynamics can be preserved. This reflects how the dynamics of the leapfrog phenomena is dependent

on the relative kinetic, H_e and vortex filament, H_s energies, as shown in **Figure 6**. It is important to note that this figure also demonstrates that the system exhibits energy loss over time, an issue that could potentially be remedied by implementing more robust operator splitting approaches and methods for calculating derivatives, as these were often calculated using lower order numerical approaches.

When scaling both the input velocity and $\hbar$ it is unclear whether the system actually changes, as $v = \hbar k$, which implies that the phase space may be unperturbed if both are scaled equally, the only thing that changes is the perceived phase velocity relative to the grid frame, which by scaling dt and $tmax$ likely returns the system to its original $\hbar = 0.1$ state. This reinforces the conclusion that the ISF model might not be physically representative below $\hbar < 0.5$ where the initial issues arise.

In determining whether the spherical Clebsch map is applicable to purely classical systems, where the minimum or dominant phase is not defined by an arbitrary parameter, a system that does not depend on the Schrödinger equation may seem preferable. [Yang et al. \(2021\)](#) implemented such a system by relying on the Mandelung transforms to relate $\rho = |\psi|^2$ and $\mathbf{u} = \langle \Delta\psi | i\psi \rangle_{\mathbb{R}}$ and inserting these into the Euler equations. Despite the loss of physical meaning for $\hbar$ they are still left with this term that quantises and defines the vorticity strength. As the paper only shows models where $\hbar \geq 0.1$, with similar parameters to Schrödinger's Smoke, it remains unclear whether the model is stable for smaller $\hbar$.

Although this still provides advantages in modelling specific classical systems where the vorticity distribution is very sharp, such that $\hbar$ can be defined as the peak, or where a minimum vorticity arises through the viscosity or size scale of the system, it still remains unclear whether these models are applicable to systems with broader distributions of vorticity. The alternative approach would be to define larger boxes with higher resolution where $\hbar$ becomes very small compared to the system. This would likely be a competitive alternative to hybrid models for simulating physical systems, but would also be computationally expensive.

5.2 Optimisation

One of the primary advantages of ISF is being able to quantise vortex filaments in a way analogous to treating them as individual particles in Lagrangian methods, whilst utilising the computational efficiency of grid based models. As seen in **Figure 4** initial tests suggest that ISF run time scales with $O(\log(x))$ with respect to resolution. However this is improbable as one of the primary bottle necks in the algorithm, the FFTW, claims to scale with $O(x \log(x))$. This result may be due to the limited parameter space tested or could potentially be caused by the oddly high run times at very low resolutions, suggesting that at lower resolutions the FFT may not always be the primary bottleneck. By expanding the tested range, the complexity scaling would likely mimic that of other fluid models with FFT being a primary determinant.

To minimise the effect of the most significant bottle necks measured, arising from the FFT and interpolation in **Algorithm 4** and **Algorithm 5** respectively. Significant runtime improvements were achieved when the manual interpolation, adopted from the Schrödinger flow paper, was substituted with its numpy alternative, `np.interpn`. The FFT bottleneck was reduced by utilising PyFFTW, which is a Python wrapper for the FFTW 3 package ([Frigo & Johnson](#)

2005). FFTW is written and heavily optimised in C which resulted in significantly faster FFTs than numpy. Further improvements were achieved by implementing multi-threading which allowed for greater use of the laptop's CPU capacity by computing FFTs in parallel. Additional minor gains were made by eliminating repeated calculations of values that remained constant throughout the simulation.

By running the code in Python, the run time compared to using a lower-level language like C++ or MATLAB will be significantly higher. However within this framework smaller optimisations and improvements to the code were possible. The code was written whilst abiding to a variety of principles including avoiding unnecessary for loops, if existing packages can complete the task faster, and utilising numpy functions rather than their native Python equivalents, as these are often optimised by running on C rather than Python. However as discussed in the previous section, Significant improvements could be made to the numerical and time integration techniques applied, these would likely increase the run time of the simulation but provide significant improvements in energy conservation and replicating physical observations.

6 CONCLUSIONS

The primary discussion of the limitations of the ISF model as $\hbar \rightarrow 0$ indicates that the model does not approach the Euler equations in this regime. The dynamics of the system can be conserved if the input velocity is scaled along with $\hbar$, however as it is unclear whether this yields an effective decrease in the minimum/quantised width of vortex filaments, further investigation is needed to test this approach. The regime in which ISF may still apply to classical systems is when the vorticity distribution is narrow and well defined, such that its quantisation, in ISF or Dr Yang's spherical Clebsch map Euler model, can be mapped to its respective distribution.

Although the calculated scaling of run time with a resolution of $O(\log(x))$ is likely the result of testing a limited parameter space, it is still clear that the ISF model outperforms its equivalent Euler equation model, even at lower relative resolutions. Further work is needed to determine a methodology that applies the benefits of spherical Clebsch maps to classical systems such that it can be applied to compete with existing Euler and Lagrange models in modelling vortex filaments in a more robust manner.

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APPENDIX A: SOURCE CODE

Inside the code used to run the ISF algorithm and generate the associated animation and Energy plots can be found in ISF_sl2014.py. An equivalent Euler simulation code can be found in EF_sl2014.py. Both follow the algorithms presented in this report along with minor addition steps, however the Euler code has some deviations from the presented algorithm where Euler equation equivalents are instead used. The code for the curve fitting of the run time is given in Curvefit_sl2014.py. Each of these python files runs independently of the other.

The simulations that are referenced or have screenshots taken of them for Figures are labeled according to their associated figure, so the .mp4 file used for Figure 3c, can be found as Figure 3c.mp4.